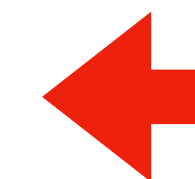
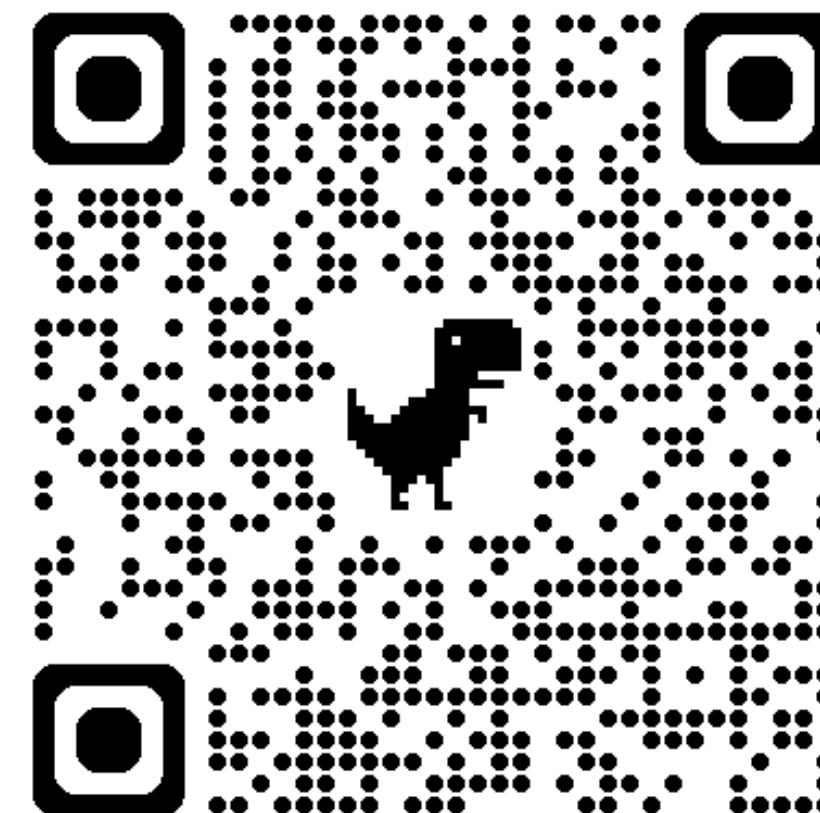


Academic Paper Writing in Artificial Intelligence

**Method Section (I):
Problem Formulation and
Model Architecture**

Instructor: Ying-Jia Lin
October 8th, 2026



Couse Website
Download slide here!

Outline

- Last week's summary (10 min)
- What the Method section is for (10 min)
- Problem Formulation in **3 steps** (30 min)
- **Your Turn (35 min, including break)**
- Model Architecture in **2 steps** (25 min)
- **Your Turn (40 min)**

Last week's summary

- Don't forget the important modules / steps in your figure.
- If your color boxes already provide clear information, you don't have to focus on the colors themselves.
- Even though you find paper's captions quite short, you should try to make your caption more detailed (longer) since we are practicing English writing. Once you know how to write longer, you definitely know how to make them shorter.

For the second item in the previous slide

Liu, Ziyi, et al. "Self-contradictory reasoning evaluation and detection." Findings of EMNLP 2024.

Jin, Qiao, et al. "Medcpt: Contrastive pre-trained transformers with large-scale pubmed search logs for zero-shot biomedical information retrieval." Bioinformatics 39.11 (2023): btad651.

Evaluation



The driver offered a ride to the teacher but he refused the offer. who is 'he'?



Since the driver is the one making the offer, **it follows that 'he' in the question refers to the driver. The teacher's refusal** of the offer does not affect who is being referred to in the initial action.



Isn't it contradictory? The reasoning says **'he' is the driver** but states it is the teacher who refused the offer. So **'he' should refer to the teacher.**

Detection



Do you think the reasoning above is self-contradictory given the question and prediction?



I couldn't see the contradiction of the reasoning. Looks fine to me.

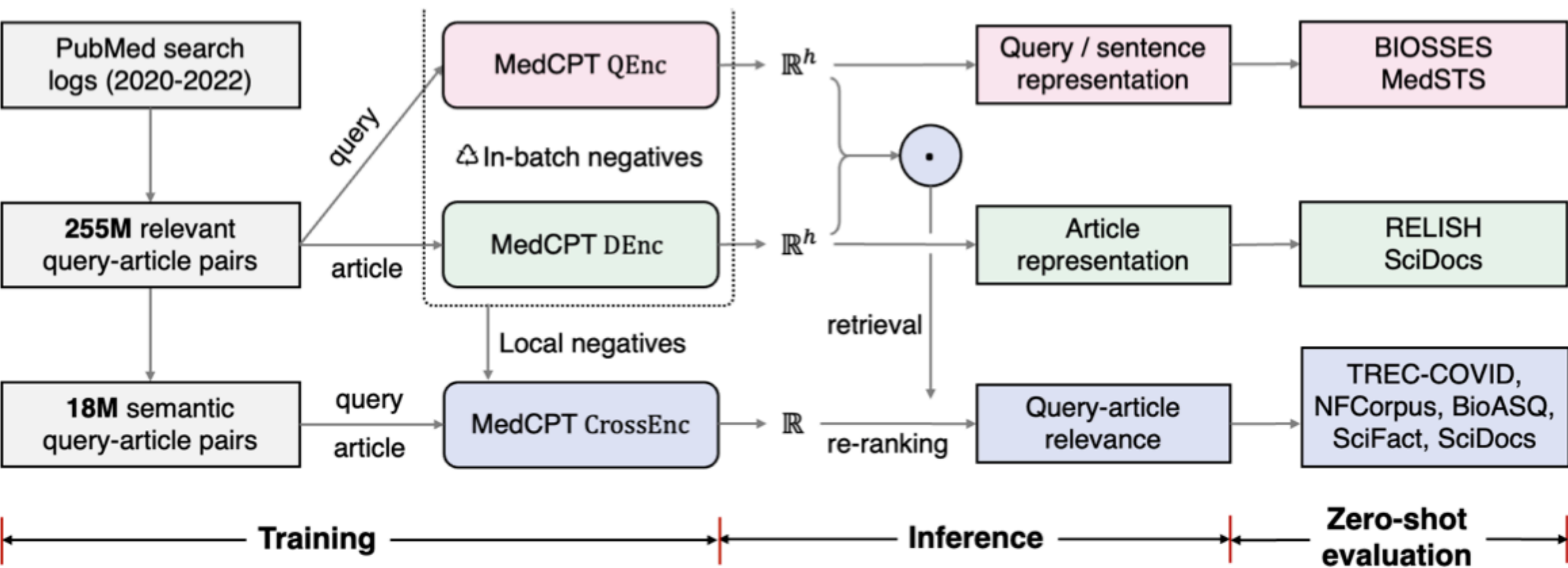


Figure 1. A high-level overview of this work. MedCPT contains a query encoder (QEnc), a document encoder (DEnc), and a cross-encoder (CrossEnc). The query encoder and document encoder compose of the MedCPT retriever, which is contrastively trained by 255M query-article pairs and in-batch negatives from PubMed logs. The cross-encoder is the MedCPT re-ranker, and is contrastively trained by 18M non-keyword query-article pairs and local negatives retrieved from the MedCPT retriever. MedCPT achieves state-of-the-art performance on various biomedical information retrieval tasks under zero-shot settings, including query-article retrieval, sentence representation, and article representation.

Figure 1: An example for self-contradictory reasoning and detection by LLMs. LLMs fail to generate consistent reasoning and are poor at detecting the self-contradiction.

Where we are

	Main Body (Your Main Paper)	
	Abstract	
	Introduction	
Depends on where you submit →	Related Work (Literature Review)	
	Methods	← Week02: Datasets as a subsection
	Results	
Depends on where you submit →	Discussion	
	Conclusion	

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	Results	↙ Week03: Metrics and Baselines are also subsections of Methods !
Depends on where you submit →	Discussion	
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Where we are

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Today we are here →	Methods	← Week02: Datasets as a subsection
	Results	← Week03: Metrics and Baselines are also subsections of Methods !
Depends on where you submit →	Discussion	
	Conclusion	

Today's Learning Objectives

- By the end of class, we hope you can:
 1. State your task in one plain sentence, then in symbols.
 2. Introduce every symbol with its meaning and its size.
 3. Say exactly what the model sees at training time and at test time.
 4. Turn your Figure 1 into the outline of the Architecture subsection.
 5. Typeset vectors, matrices, sets, and function names correctly.

What the Method section is for

- A reader should be able to **rebuild (or reproduce) your method** from this section **alone**.
- Not "understand roughly what you did". Rebuild.
 - That is a high bar, and it is why the Method section is long.
 - It is also why we split it over two weeks.

Don't be afraid !!

經典台式紅燒牛肉麵

準備食材 (約 4 人份)

- 肉與蔬菜：牛腱肉 800g (約兩條)、牛番茄 2 顆、白蘿蔔 1 條、紅蘿蔔 1 條
- 辛香料：青蔥 2 支、薑 1 小塊 (切片)、蒜頭 5 瓣、辣椒 1 根、市售牛肉麵滷包 1 個
- 靈魂調味：辣豆瓣醬 2 大匙、醬油 120ml、米酒 50ml、冰糖 1 大匙、水 1500ml
- 上桌配料：麵條 4 份、青江菜 4 把、酸菜適量、蔥花少許

料理步驟

1. 川燙去腥：將牛腱切大塊 (約 4-5 公分，煮熟會縮水)。備一鍋冷水，放入牛腱與少許薑片，開中火煮滾後撈起，用清水洗淨表面浮沫雜質備用。
2. 爆香炒底：熱鍋下少許油，放入蔥段、薑片、蒜頭與辣椒爆香，接著加入辣豆瓣醬，用中小火炒出紅油與醬香。
3. 翻炒上色：將牛腱肉下鍋拌炒，加入冰糖炒至微融化，接著沿著鍋邊燴入米酒與醬油，讓牛肉均勻沾附醬汁並炒出香氣。
4. 慢火精燉：放入切塊的番茄、紅、白蘿蔔，倒入清水 (需淹過所有食材) 並投入滷包。大火煮滾後轉小火，加蓋慢燉約 1.5 至 2 小時，直到牛肉用筷子能輕鬆戳透。
5. 煮麵組裝：另起一鍋滾水，將麵條與青江菜燙熟後撈起裝碗。鋪上牛肉塊與蘿蔔，淋上熱騰騰的濃郁原湯，最後點綴蔥花與酸菜即可上桌！

- As we mentioned earlier, **Datasets** is a subsection in **Methods**.

- Writing content in Methods is just like writing the ingredient part of a recipe.

- ➡ Imagine you are writing something to help others **cook (reproduce) a bowl of beef noodles (your method)**

✗ some beef

✓ 800g beef shank (two pieces)

✗ We use most of the qualified questionnaire responses.

✓ We collected 412 responses and excluded 37 that were incomplete, leaving 375 for the analysis.

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We will have two subsections today.

Problem Formulation

Model Architecture

Problem Formulation

Similar Names of Problem Formulation

Problem Formulation	Background	Task Introduction	(Empty Subsection Title)
Problem Statement	Preliminaries	Task Definition	Overview
Problem Setup		Formal Definition	

In AI/CS papers, this subsection is usually the first section with many **math symbols**.

Example 1 for Problem Formulation (1/2)

Kim, Yoon, and Alexander M. Rush. "Sequence-level knowledge distillation." EMNLP 2016.

2 Background

2.1 Sequence-to-Sequence with Attention

Let $\mathbf{s} = [s_1, \dots, s_I]$ and $\mathbf{t} = [t_1, \dots, t_J]$ be (random variable sequences representing) the source/target sentence, with I and J respectively being the source/target lengths. Machine translation involves finding the most probable target sentence given the source:

$$\operatorname{argmax}_{\mathbf{t} \in \mathcal{T}} p(\mathbf{t} | \mathbf{s})$$

where $\mathcal{T}$ is the set of all possible sequences. NMT models parameterize $p(\mathbf{t} | \mathbf{s})$ with an *encoder* neural network which reads the source sentence and a *decoder* neural network which produces a distribution over the target sentence (one word at a time) given the source. We employ the attentional architecture from Luong et al. (2015), which achieved state-of-the-art results on English $\rightarrow$ German translation.³

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Objective of this subsection:

- Define the math symbols
- Tell readers what the target task in your paper is

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Example 1 for Problem Formulation (2/2)

Kim, Yoon, and Alexander M. Rush. "Sequence-level knowledge distillation." EMNLP 2016.

The section is called background ...

Concretely, assume we are learning a multi-class classifier over a data set of examples of the form (x, y) with possible classes $\mathcal{V}$. The usual training criteria is to minimize NLL for each example from the training data,

$$\mathcal{L}_{\text{NLL}}(\theta) = - \sum_{k=1}^{|\mathcal{V}|} \mathbb{1}\{y = k\} \log p(y = k \mid x; \theta)$$

where $\mathbb{1}\{\cdot\}$ is the indicator function and p the distribution from our model (parameterized by θ).

This objective can be seen as minimizing the cross-entropy between the degenerate data distribution (which has all of its probability mass on one class) and the model distribution $p(y \mid x; \theta)$.

Example 1 for Problem Formulation (2/2)

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← What the model is trained with

← Naming convention (later the authors will write something like this)

Example 2 for Problem Formulation

(Empty tile case)

Lewis, Patrick, et al. "Retrieval-augmented generation for knowledge-intensive nlp tasks." NeurIPS 2020.

2 Methods

We explore RAG models, which use the input sequence x to retrieve text documents z and use them as additional context when generating the target sequence y . As shown in Figure 1, our models leverage two components: (i) a retriever $p_\eta(z|x)$ with parameters η that returns (top-K truncated) distributions over text passages given a query x and (ii) a generator $p_\theta(y_i|x, z, y_{1:i-1})$ parametrized by θ that generates a current token based on a context of the previous $i - 1$ tokens $y_{1:i-1}$, the original input x and a retrieved passage z .

To train the retriever and generator end-to-end, we treat the retrieved document as a latent variable. We propose two models that marginalize over the latent documents in different ways to produce a distribution over generated text. In one approach, *RAG-Sequence*, the model uses the same document to predict each target token. The second approach, *RAG-Token*, can predict each target token based on a different document. In the following, we formally introduce both models and then describe the p_η and p_θ components, as well as the training and decoding procedure.

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A small tip to save space

✓ A retriever $p_\eta(z | x)$ with parameters η ...

⚠ We use $p_\eta(z | x)$ with η ..., where $p_\eta(z | x)$ is a retriever, and η is the number of parameters.

How to write a Problem Formulation section?

Math Notations

- ✓ Each sequence is tokenized in to a list of tokens $x = \{x_0, x_1, \dots, x_{n-1}\}$, where n is the input sequence length.

Task Definition

- ✓ Our goal is to learn mapping functions between two domains X and Y , given training examples $\{x_i\}_{i=1}^N$ and $\{y_j\}_{j=1}^M$.

How to write a Problem Formulation section?

Constraints

Conway A, Hada A, Schluter J, Xu H, Lowden D, Miller T, Donald K, Teodorczuk A AI-Based Structured Information Extraction From Synthetic Nursing Handover Transcripts: Comparative Evaluation of Large Language Models. JMIR Nursing 2026;9:e106133 doi: 10.2196/106133

✅ This study used a model-evaluation design, corresponding to the “LLM evaluation” research-design category in TRIPOD-LLM (Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis–Large Language Models) [16] (Checklist 1).

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💡 Central guide: Formally define the **inputs**, expected **outputs**, and **research boundaries** of your task.

How to write a Problem Formulation section?

Learning Objective

See, Abigail, Peter J. Liu, and Christopher D. Manning. "Get to the point: Summarization with pointer-generator networks." ACL 2017.

During training, the loss for timestep t is the negative log likelihood of the target word w_t^* for that timestep:

$$\text{loss}_t = -\log P(w_t^*) \quad (6)$$

and the overall loss for the whole sequence is:

$$\text{loss} = \frac{1}{T} \sum_{t=0}^T \text{loss}_t \quad (7)$$

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💡 Optional: Include a high-level objective if it helps **define the task goal**. If your **Loss Function depends on model internals**, defer it to the **Method section**.

Sentence Bank - Problem Formulation

- We study the task of X : given A , the goal is to produce B .
- Formally, let A denote ... and B denote ...
- We denote the training set by $\mathcal{D} = (x_i, y_i)_{i=1}^N$.
- At inference the model observes only A .
- “Throughout the paper, bold lowercase letters denote vectors and bold uppercase letters denote matrices.”  One line only throughout the paper.
- We use i to index reports and t to index tokens.

Your Turn

Model Architecture

The overview figure is your outline.

- Your Figure 1 already contains the structure of this subsection.
 - one box in the figure → one paragraph
 - the order of the arrows → the order of the paragraphs
 - the name in the box → the name in the paragraph heading
- If the figure and the subsection disagree about the order, one of them is wrong. This is why we drew the figure first.

The two steps of a Model Architecture section

1 Announce — name the components, and point at the figure.

2 One paragraph per component — what goes in, what happens, what comes out.

 **Mark what is standard** — cite it and move on, so the reader knows where to look.

 **Target length:** as long as it needs to be. This is the longest subsection in your paper.

Step 1: announce and point at the figure

"As shown in Figure~\ref{fig:overview}, \method has three components: a report encoder, a cluster-based scorer, and a ranking head. We describe each in turn."

Three things this sentence does:

- tells the reader how many parts there are, so they can keep count
- gives the order you will use
- ties the text to the figure with a real \ref

Step 2: One paragraph per component

For example, if my method has
three components:

(1) Segment Model

(2) Classifier

(3) Proposed Framework

how to separate the words in a sequence. In other words, the segment model acts as a teacher for the classifier. The following sections first introduce the details of the two modules and then describe the training process in our proposed framework.

2.1 Segment Model

In our approach, we employ the Segment Language Model (SLM) proposed by Sun and Deng (2018) as our segment model for providing pseudo-labels in our proposed framework. SLM is a language model which segments a sequence according to the probability of generating <eos> (*end of a segment*) as the next character. For example, given a sequence $\{x_1, x_2, x_3\}$, two segments $\{x_1, x_2\}$ and $\{x_3\}$ can be obtained via SLM if the probability of generating <eos> after x_2 is higher than that after x_3 . We omit an exhaustive unsupervised training process of SLM (Sun and Deng, 2018) and refer readers to the original paper for more details.

2.2 Classifier

We follow Xue (2003) to treat CWS as a sequence tagging task. As the illustration in Figure 1 (b), we add fully-connected layers (FC) on top of the BERT (Devlin et al., 2018) model as our classifier:

$$p_{ij} = \text{softmax}(W \cdot h_{ij} + b), \\ \forall i \in [1, N], \forall j \in [1, T_i],$$

where p_{ij} is the probability of the j -th token in the i -th sequence, N is the number of examples in a dataset, and T_i is the length of the i -th sequence. $W \in \mathbb{R}^{d \times k}$ and $b \in \mathbb{R}^k$ are trainable parameters for k output tagging labels, and $h_{ij} \in \mathbb{R}^d$ is the output hidden state of BERT at the j -th token with d dimensions.

2.3 Training Framework

Our training framework is composed of two stages, as illustrated in Figure 1. In the first stage, we follow Sun and Deng (2018) to train the segment model. We then perform pseudo-labeling and create high-quality word segments with the segment model. In the second stage, we use these pseudo-labeled segments to train the classifier with cross-entropy loss:

$$\mathcal{L} = - \sum_{i=1}^N \sum_{j=1}^{T_i} y_{ij} \log(p_{ij})$$

where y_{ij} is the pseudo-label of the j -th token in the i -th sequence. We adopt the tagging schema with binary tagging labels ($k = 2$), where "1" represents "segment from the next character" and "0" indicates "do not segment." The results with other tagging schemas are included in Appendix A.1.

Our goal is to provide the pre-trained classifier with pseudo-labels as training data, allowing it to utilize the knowledge acquired during its pre-training stage to learn further and improve its performance on Chinese word segmentation.

3 Experiments

3.1 Datasets

We conduct experiments on the eight widely used datasets, including AS, CityU, MSR and PKU from the SIGHAN 2005 Bakeoff (Emerson, 2005), SXU from the SIGHAN 2008 Bakeoff (Jin and Chen, 2008), CNC and CTB from the Penn Treebank (Xue et al., 2005), and UD from the CoNLL 2018 Shared Task (Zeman et al., 2018). We use the evaluation script provided by SIGHAN 2005 (Emerson, 2005) and report the results in F1 scores.

Step 2: One paragraph per component

For example, if my method has
two sub-components:

(1) *Prompt Design*

(2) *RAG*

Retriever Training Set Construction

Since our dataset lacked direct AIS annotations for specific text spans, we adopted a proxy task strategy to train the retriever. We used trauma-related *ICD-10-CM* codes (S00 to T75) and their official descriptions to create training pairs, allowing the retriever to learn semantic alignments between clinical narratives and structured injury codes. At this phase, 8 records with no *ICD* (*International Classification of Diseases*) codes were removed from the 4092 cases. To mitigate the risk of overfitting the retriever on *ICD-10-CM* descriptions, we randomly selected 70% (2859 discharge diagnoses) of the records from the pool of 4084 cases for training rather than using all available cases. The remaining 1225 (30%) records were not used for training or hyperparameter tuning. The 4 supplementary test cases (mentioned in the previous section) were drawn from the held-out portion of this pool. No case in the final test set appears in the retriever's training data. As a single discharge diagnosis may contain multiple *ICD-10* codes, this process yielded 7570 training examples in total. The system's final performance was evaluated exclusively on a held-out test set of 100 discharge diagnoses. These cases were annotated by a physician, with a subset independently reviewed by a senior trauma specialist for quality assurance.

Using LLMs for ISS Calculation

Prompt Design

In our prompts, we instructed an LLM to take the role of a senior trauma registrar. We then guided an LLM to extract the AIS codes based on the AIS 1998 code set and explicitly described how to calculate the ISS score from the extracted AIS codes. In addition to the guidance on ISS calculation, we also provided

2 simple examples to enhance the LLM's understanding of ISS calculation. The prompts also included requirements to mention explanations or uncertainties to improve performance via reasoning [17] and to provide more transparent responses. Each model was instructed to return its analysis in JSON format, containing a required "explanation" field describing how the AIS scores and ISS were derived, a required "ISS" field holding the calculated score, and optional "limitations" and "uncertainties" fields. The final ISS used in all analyses was taken from the "ISS" field of each response; in the few cases where a model returned this field as an arithmetic expression rather than a single integer, the resulting value of that expression was used, and the ISS was never read from the "explanation" text. The same base prompt was used for all evaluated LLMs and experimental settings, with the retrieved AIS entries inserted before the discharge diagnosis in the RAG settings. The prompts used in this study are provided in section A in [Multimedia Appendix 1](#).

RAG

While LLMs possess broad knowledge, they often lack specialized information such as the AIS 1998 code set. When prompted directly, these models exhibit a tendency to hallucinate or generate factually incorrect AIS codes, thereby making wrong ISS calculations. To mitigate this, we used a RAG framework to ground the model's responses in an authoritative knowledge source. Our RAG architecture consists of 2 core components: a retriever and a generator. The retriever searches a vector database built from the AIS 1998 dictionary, which contains 1366 entries. Each entry has a 6-digit AIS predot code, a free-text injury description, an AIS severity value, and an ISS body region. For example, entry 150200 has the description

Step 2: Input shape and output shape

2.2 Classifier

We follow [Xue \(2003\)](#) to treat CWS as a sequence tagging task. As the illustration in Figure 1 (b), we add fully-connected layers (FC) on top of the BERT ([Devlin et al., 2018](#)) model as our classifier:

$$p_{ij} = \text{softmax}(W \cdot h_{ij} + b), \\ \forall i \in [1, N], \forall j \in [1, T_i],$$

where p_{ij} is the probability of the j -th token in the i -th sequence, N is the number of examples in a dataset, and T_i is the length of the i -th sequence. $W \in \mathbb{R}^{d \times k}$ and $b \in \mathbb{R}^k$ are trainable parameters for k output tagging labels, and $h_{ij} \in \mathbb{R}^d$ is the output hidden state of BERT at the j -th token with d dimensions.

💡 For AI/CS papers, writing shape information will help readers understand your method more clearly .

$$\begin{array}{ccccc} [x_1 & x_2 & x_3] & \times & \begin{bmatrix} w_{1,1} & w_{1,2} \\ w_{2,1} & w_{2,2} \\ w_{3,1} & w_{3,2} \end{bmatrix} & + & [b_1 & b_2] & = & [o_1 & o_2] \\ \mathbb{R}^{1 \times 3} & & & & \mathbb{R}^{3 \times 2} & & \mathbb{R}^{1 \times 2} & & & \mathbb{R}^{1 \times 2} \\ \text{Input} & & & & \text{Internal computation} & & \text{Output} & & & \end{array}$$

Mark what is standard

"We use a standard Transformer encoder \citep{vaswani2017}; we describe only the parts we change."

This does three things at once:

- Saves space — you do not re-explain attention in 2026.
- Directs attention — the reader now knows where your contribution is.
- Is honest — it says clearly what you did not invent.

Same rule as the figure last week: detail belongs where your idea is.

What does not belong to this subsection

Save all of this for Week 6:

- loss functions, and how they are combined
- the training algorithm or any pseudocode
- optimiser, learning rate, batch size, number of epochs
- hardware, training time, random seeds

Today: what the model is. Next week: how it is trained.

Sentence Bank

- "As shown in Figure~\ref{fig:overview}, \method has three components: ..."
- "Report encoder. The encoder takes X and returns Y."
- "We use a standard X \citep{...}; we describe only our modifications."
- "The output of this component is passed to the ranking head (Section~\ref{sec:rank})."
- "Unless stated otherwise, all layers use hidden sized."
- "This is the only component we introduce; the rest of the pipeline is unchanged."

Common mistakes in the Method section

- Describing your code instead of your method (class names, file names, library calls)
- A wall of text with no bold component names
- Explaining a standard well-known model (e.g., Transformer) for half a page
- Components in a different order from the figure
- No output shapes (in AI/CS papers)

Your Turn

Zotero

NEXT WEEK ATTENTION!!

- We will have “Algorithms, Loss Functions, and Training Details” next week.
- Please bring your Methods materials to class!
 - **Two weeks** for the method section.
 - **We will have more time for revisions next week.**

Thank you for your participation 